HARD · ADVANCED MATH · ANSWER KEY

Advanced Math

14

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 C

C. -25

Choice C is correct. It's given that the equation $-9x^2 + 30x + c = 0$ has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $-4ac + b^2$, is equal to zero. It follows that for the given equation, $a = -9$ and $b = 30$. Substituting -9 for a and 30 for b into $b^2 - 4ac$ yields $30^2 - 4 \cdot 9c$, or $900 + 36c$. Since the discriminant must equal zero, $900 + 36c = 0$. Subtracting $36c$ from both sides of this equation yields $900 = -36c$. Dividing each side of this equation by -36 yields $-25 = c$. Therefore, the value of c is -25.

Choice A is incorrect. If the value of c is 3, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice B is incorrect. If the value of c is 0, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice D is incorrect. If the value of c is -53, this would yield a discriminant that is less than zero. Therefore, the given equation would have no real solutions, rather than exactly one solution.

Q2 A

A. 8

Choice A is correct. It's given that the function P models the population, in thousands, of a certain city t years after 2003. The value of the base of the given exponential function, 1.04, corresponds to an increase of 4% for every increase of 1 in the exponent, $\frac{6}{4}t$. If the exponent is equal to 0, then $\frac{6}{4}t = 0$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = 0$. If the exponent is equal to 1, then $\frac{6}{4}t = 1$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = \frac{4}{6}$, or $t = \frac{2}{3}$. Therefore, the population is predicted to increase by 4% every $\frac{2}{3}$ of a year. It's given that the population is predicted to increase by 4% every n months. Since there are 12 months in a year, $\frac{2}{3}$ of a year is equivalent to $\frac{2}{3}12$, or 8, months. Therefore, the value of n is 8.

Choice B is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice C is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{4}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice D is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{1}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Q3 C

C. $w = \frac{x^2}{y} - 19$

Choice C is correct. Dividing each side of the given equation by 2 yields $\frac{14x}{14y} = \frac{2\sqrt{w+19}}{2}$, or $\frac{x}{y} = \sqrt{w+19}$. Because it's given that each of the variables is positive, squaring each side of this equation yields the equivalent equation $\frac{x^2}{y} = w + 19$. Subtracting 19 from each side of this equation yields $\frac{x^2}{y} - 19 = w$, or $w = \frac{x^2}{y} - 19$.

Choice A is incorrect. This equation isn't equivalent to the given equation.

Choice B is incorrect. This equation isn't equivalent to the given equation.

Choice D is incorrect. This equation isn't equivalent to the given equation.

Q4

D

D. $rm = 90{,}0001.06^{\frac{m}{12}}$

Choice D is correct. It's given that the function p models the population of Lowell t years after a census. Since there are 12 months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $pt = 90{,}0001.06^{t}$ yields $p\frac{m}{12} = 90{,}0001.06^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $rm = 90{,}0001.06^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5

-1

The correct answer is -1. A quadratic equation in the form $ax^2 + bx + c = 0$, where a, b, and c are constants, has exactly one solution when its discriminant, $b^2 - 4ac$, is equal to 0. In the given equation, $-16x^2 - 8x + c = 0$, $a = -16$ and $b = -8$. Substituting -16 for a and -8 for b in $b^2 - 4ac$ yields $-8^2 - 4{-16}c$, or $64 + 64c$. Since the given equation has exactly one solution, $64 + 64c = 0$. Subtracting 64 from both sides of this equation yields $64c = -64$. Dividing both sides of this equation by 64 yields $c = -1$. Therefore, the value of c is -1.

Q6

D

D. $C = 402.9^t$

Choice D is correct. It's given that an exponential model estimates that the number of comments on an article increased by a fixed percentage at the end of each hour. Therefore, the model can be represented by an exponential equation of the form $C = Ka^t$, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and K and a are constants. It's also given that when the article was first featured on the home page of the news website, there were 40 comments on the article. This means that when $t = 0$, $C = 40$. Substituting 0 for t and 40 for C in the equation $C = Ka^t$ yields $40 = Ka^0$, or $40 = K$. It's also given that the number of comments on the article at the end of an hour had increased by 190% of the number of comments on the article at the end of the previous hour. Multiplying the percent increase by the number of comments on the article at the end of the previous hour yields the number of estimated additional comments the article has on its home page: $40\frac{190}{100}$, or 76 comments. Thus, the estimated number of comments for the following hour is the sum of the comments from the end of the previous hour and the number of additional comments, which is $40 + 76$, or 116. This means that when $t = 1$, $C = 116$. Substituting 1 for t, 116 for C, and 40 for K in the equation $C = Ka^t$ yields $116 = 40a^1$, or $116 = 40a$. Dividing both sides of this equation by 40 yields $2.9 = a$. Substituting 40 for K and 2.9 for a in the equation $C = Ka^t$ yields $C = 402.9^t$. Thus, the equation that best represents this model is $C = 40(2.9)^t$.

Choice A is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 19% of the number of comments at the end of the previous hour, rather than 190%.

Choice B is incorrect. This model represents a situation where the number of comments at the end of each hour increased by 90% of the number of comments at the end of the previous hour, rather than 190%.

Choice C is incorrect. This model represents a situation where the number of comments at the end of each hour was 19 times the number of comments at the end of the previous hour, rather than increasing by 190% of the number of comments at the end of the previous hour.

Q7

D

D. Zero

Choice D is correct. Any quantity that is positive or negative in value has a positive value when squared. Therefore, the left-hand side of the given equation is either positive or zero for any value of x. Since the right-hand side of the given equation is negative, there is no value of x for which the given equation is true. Thus, the number of distinct real solutions for the given equation is zero.

Choices A, B, and C are incorrect and may result from conceptual errors.

Q8

-12

The correct answer is -12. It's given that function f is a quadratic function where $f(-20)=0$ and $f(-4)=0$. It follows that the graph of $y=f(x)$ in the xy-plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the x-coordinate of the vertex of the graph is the average of a and b. Therefore, the x-coordinate of the vertex of the graph of $y=f(x)$ is $\frac{-20+-4}{2}$, or -12. It's given that the graph of $y=f(x)$ in the xy-plane has a vertex at $(r, -64)$. It follows that the value of r is -12.

Q9 either 2 or 8

The correct answer is either 2 or 8. Substituting $x=a$ in the definitions for f and g gives $f(a)=-\frac{1}{2}(a-4)^2+10$ and $g(a)=-a+10$, respectively. If $f(a)=g(a)$, then $-\frac{1}{2}(a-4)^2+10=-a+10$. Subtracting 10 from both sides of this equation gives $-\frac{1}{2}(a-4)^2=-a$. Multiplying both sides by -2 gives $(a-4)^2=2a$. Expanding $(a-4)^2$ gives $a^2-8a+16=2a$. Combining the like terms on one side of the equation gives $a^2-10a+16=0$. One way to solve this equation is to factor $a^2-10a+16$ by identifying two numbers with a sum of -10 and a product of 16. These numbers are -2 and -8, so the quadratic equation can be factored as $(a-2)(a-8)=0$. Therefore, the possible values of a are either 2 or 8. Note that 2 and 8 are examples of ways to enter a correct answer.

Alternate approach: Graphically, the condition $f(a)=g(a)$ implies the graphs of the functions $y=f(x)$ and $y=g(x)$ intersect at $x=a$. The graph $y=f(x)$ is given, and the graph of $y=g(x)$ may be sketched as a line with y-intercept 10 and a slope of -1 (taking care to note the different scales on each axis). These two graphs intersect at $x=2$ and $x=8$.

Q10 5

The correct answer is 5. It's given that $fx = -a^x+b$. Substituting $-a^x+b$ for fx in the equation $y=fx-15$ yields $y=-a^x+b-15$. It's given that the y-intercept of the graph of $y=fx-15$ is $0, -\frac{99}{7}$. Substituting 0 for x and $-\frac{99}{7}$ for y in the equation $y=-a^x+b-15$ yields $-\frac{99}{7}=-a^0+b-15$, which is equivalent to $-\frac{99}{7}=-1+b-15$, or $-\frac{99}{7}=b-16$. Adding 16 to both sides of this equation yields $\frac{13}{7}=b$. It's given that the product of a and b is $\frac{65}{7}$, or $ab=\frac{65}{7}$. Substituting $\frac{13}{7}$ for b in this equation yields $a\frac{13}{7}=\frac{65}{7}$. Dividing both sides of this equation by $\frac{13}{7}$ yields $a=5$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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